CHAPTER 1

GENERAL PRINCIPLES

Material handling plays an important part in industrial economy. Great steam plants would not be possible without coal-handling equipment, nor could steel, paper, chemical, cement, and manufacturing plants function without conveyors. Mass production in industry depends on the conveying system, and expansions in mass production invariably follow improvements in methods of handling the materials involved.

Tricky Materials. Most performance disappointments result from a failure on the part of the engineer to make a comprehensive study of the material and its reaction in a conveyor. Tricky materials are many; it is worth while to list a few of them.

Carbon-black pellets are extremely fragile, and if broken up they are useless for certain purposes. Gentle handling is essential, and any conveyor having a grinding action will be a failure.

Sulphur develops static electricity through friction, and when discharged along a chute it builds up a voltage later equalized by spark discharge. The disasters resulting from chuting sulphur into the hold of a vessel about to sail can readily be appreciated. Chutes must be lined with aluminum or aluminum bronze, and the conveying system must be such that the static charge will be dissipated harmlessly.

In certain brewing processes a mixture of ground malt and hulls has been found best in the vats. If the hulls are not ground up the liquor filters freely. If the hulls are ground up the whole process slows down, and the output of the plant is reduced. Special grinding machines have been developed which do not pulverize the hulls, but if this prepared mixture then is handled by a high-speed elevator, by a scraper conveyor, or even by a screw conveyor, the hulls suffer degradation and the processing is slow and costly.

Cement is an odd material, often causing the designer much grief. The capacity of a bucket elevator for handling cement is much less than its tabulated value. Air is trapped in the buckets as the cement enters. A few vents in the floor of each bucket will remedy this trouble to some extent, but oversize buckets should be selected Another curious characteristic is that, while aerated, cement flows freely, but after it has stood awhile in storage or in a car the air is squeezed out, and it will not flow.

Soda ash is even more lively when aerated. The author has seen soda ash freshly stored by pneumatic conveyor flow out of an opening in a bin in a way which approached the flow of water; that is, its angle of repose was almost zero. It has a troublesome habit of sticking to the buckets of a centrifugal discharge elevator and discharging back down the return leg, though it handles nicely in a slow-speed continuous bucket elevator. Since the dust has an irritating effect on the mucous membrane, dustless operation is essential. For unloading bulk soda ash from box cars the pneumatic conveyor, which functions like a vacuum cleaner, has advantages.

Engineers designing sewage-treatment plants have run into trouble in the selection of elevators for handling wet sludge. Conveying en masse fails utterly, as the sludge builds up on the walls of the duct and on the impellers, completely jamming the machine. It quickly chokes a bucket elevator by adhering to the buckets, and it contains a fine grit which soon cuts through the chain joints and sprocket wheels. Handling sludge is one of the most difficult conveying problems.

Fuller's earth in granular form, granulated slate for prepared roofing, granulated sugar, table salt, and the like must not be ground up or degraded. Food products, talc, face powder, tooth paste, etc., must not suffer discoloration or come in contact with lubricants. The shell or protective sheath of oats must not be damaged by rough handling. Some coals when damp are very corrosive to steel plate. Flaxseed at high velocity acts almost like a sand blast. Copra, under the high-velocity action of a pneumatic conveyor, may yield up its oil content, causing difficult deposits in the centrifugal receiver and reducing the quality of the copra. Ashes are abrasive, dusty, and troublesome when dry and corrosive when wet.

Borax, ground feldspar, and sand are highly abrasive. Aluminum oxide in finely powdered form has a cutting action just a little lower than diamond dust. If it gets into the chain joints, as it has an uncanny ability to do, the chain is soon ruined.

These are only a few of the tricky materials—the list is long. Table 1 mentions bulk materials that frequently enter into material-handling problems, with comments on their action when handled in conveyors.

Dust Explosions and Dust Prevention. With the extensive use of materials having an affinity for oxygen there is a growing dust hazard in connection with materials handling installations. The subject is covered at length in Chapter 22. All mechanical conveyors can be housed in so that dust will not be scattered; for example, a cover

plate may be placed over the carrying run of a flight conveyor or upon the trough of a screw conveyor. Continuous-flow conveyors are inherently dustless. Belt conveyors with trippers are difficult; they are discussed in Chapter 10. Pneumatic conveyors are dustless except at the discharge from the receiver or dust collector, but there is an internal risk that a fragment of steel in a finely divided combustible material might strike a spark by impact against the wall of the duct.

Plant Layout and Conveyor Layout. The type of conveyor is only one part of the layout of a plant. There may be, and usually are, several equally effective types of material-handling equipment, but the flow of material through the plant — that is to say the *engineering* of the conveying system — has a far-reaching effect on the cost of production. Savings or losses in every department may depend on the material-handling scheme. Too often first cost receives more attention than reliability. As a rule it takes only two or three breakdowns of a badly engineered conveying system to wipe out the difference in cost for the best machine that might have been installed.

Price vs. Quality. The responsible conveyor manufacturer with a reputation to maintain will not cheapen his product to compete with something he knows to be inferior. It is not difficult to find an assembled skip hoist, drag scraper, etc., cheaper than the product of a first-class manufacturer, since the drum, sheaves, shaft, and bucket can be purchased here and there from small shops which at the moment need work. If the assembled equipment fails, the seller may sidestep responsibility. The subcontractor has been paid and is not inclined to lose a few dollars he made on a part whose application he knew nothing about. The user holds the bag, as under the circumstances he should. His headache can be avoided if he weighs reputation against price differential.

States and municipalities must purchase from the lowest bidder, and maintenance guarantees are rarely required. Maintenance costs are usually lower in the privately owned plant because, for the sake of his own reputation, the works engineer as a rule buys the best equipment. Two installations stand out in the author's memory. In 1907 a pivoted bucket carrier was purchased by an Illinois institution on a mandatory competitive basis. It handled about 12,000 tons of coal and 1000 tons of ashes per year. The cost of maintenance was high, and finally in 1938 the equipment was in such generally bad condition that bids were asked for a practically complete replacement. It so happened that in 1908 a carrier costing about 20 per cent more, almost identical in size, layout, and tonnage handled, was installed

in a neighboring manufacturing plant. In 1938 it was still in fine condition with a maintenance cost averaging $50 per year (material only).

Engineering Responsibility. Considerations of suitability are of even greater importance to the consulting engineer than to the manufacturer or contractor, whose guarantee may be limited to quality or capacity. On one occasion it was required to unload slate granules from box cars to an overhead bin at 40 tons per hour. This slate was a clean sharp product, screened to about ¼-in. cubes, used for facing roofing paper. The consulting engineers specified a pneumatic system, with the usual guarantees of quality and capacity. When the equipment was placed in operation it was immediately apparent that the high velocity, the jostling in the nozzle and flexible hose, and the impact at the elbows so degraded and rounded the particles that the slate was unusable. The contractor sued for the 10 per cent balance due him, and the purchasers sued for the return of the 90 per cent already paid. Quality and capacity were acknowledged. The court held that the responsibility rested with the consulting engineer who specified the type of equipment and who should have studied the manufacturing requirements; that it was not the responsibility of the contractor to determine whether the pneumatic system specified by the engineer would make the material unsuitable. He was required merely to unload the material at 40 tons per hour, and he did so. Eventually a set of power shovels was installed, with a belt conveyor alongside the track, and a belt and bucket elevator, which functioned with very little damage to the granules.

Capacity. "Handling capacity" may have different meanings to different persons. The engineer must be specific in his statements. If it is desired to unload ten 50-ton cars in 10 hours, a conveyor capacity of 50 tons per hour will not suffice. If an unloading capacity of 100 tons per hour is specified, that does not mean that two 50-ton cars will be unloaded, because time is lost in spotting the cars, or if power shovels are involved the car-door bulkheads must be removed. If a 50-ton-per-hour pneumatic unloader is used the cleanup of the last few tons in each car is done at a fraction of that rate, and then the emptied car must be moved out and a loaded car moved in.

Samples Important. It is a good idea to save a sample of the material for which you are asked to specify a conveyor. Memory sometimes is amazingly short when things do not work out exactly as expected by the purchaser. In a pneumatic system, for instance, if the material proves to have different characteristics from those for which the velocity and volume of air were determined, the results may be quite different from expectations.

Importance of Clearance around Machinery. One of the most irritating circumstances is lack of comfortable headroom and clearance around machinery. An overhead girder, or a pipe which scrapes the head, or scant lateral clear space which results in a smear on the clothes, is a defect in design. It may be a serious factor should a man be injured and bring suit for damages.

Importance of Engineering. It is often a temptation to use a standard machine for a job which seems to call for no engineering. It is true that by so doing the engineering cost is eliminated, but the engineering cost may be only a small part of the total cost, and the purchaser should carefully consider whether the all-round efficiency of his installation is best assured by having standardized equipment intended to fulfill average requirements. It may be that such a machine will do the job, but if it does not the saving in engineering cost soon evaporates.

Each manufacturer maintains a staff of specialists, skilled through years of experience. The purchaser or engineer who develops a project without seeking their cooperation may let himself in for serious troubles and large losses. The engineering of material-handling installations is not always as elementary as it appears to be.

Liberality of Design. Possibly this designation requires an explanation. When a specification calls for a certain capacity the contractor receives no additional compensation, except a sense of pleasure, if that capacity is exceeded. He may be in serious difficulties if his installation fails even by a small margin to meet the guarantee. This is especially true in government work where the inspector has no discretion as regards approval.

A coal bunker may be specified for 500-ton capacity figured at 50 lb. per cu. ft.; but coal often weighs less than 50 lb. per cu. ft., and if it does the capacity is too small. The angle of repose of a material may have been assumed to be 35° and then may be found to be 45° when the material is deposited in a container with a cone top, and so there is a vacant space at the top of the container. The capacity cannot easily be increased. The handling capacity of conveyors is definitely specified, but many materials weigh less than expected, owing to aeration, and the conveyor is not quite up to capacity. Play safe and leave a margin in your design to take care of contingencies. Your final check will come along sooner.

Specifications. The specifications of consulting or project engineers are nearly always fair to both parties of a contract, but it is advantageous for the bidder to read through the "small print." This is true particularly with government specifications, which sometimes have odd angles. On one contract including a coal-storage structure adja-

cent to the boiler house and a larry runway from beneath the bin through the boiler house, the author completed the bin structure and notified the government engineer that, as soon as the boiler house framework was erected so that the larry track could be suspended therefrom, the job could be wound up at once. The contract carried a penalty of $100 per day. Two months later the government advised that the work could be finished and a week later the job was wound up and accepted. In due course final payment, less $6000 penalty, was received. The engineer agreed that the contractor could not hang the larry runway from sky hooks but pointed to a clause in " small print " which required the contractor to ask for an extension of time should his work be interrupted, and the Comptroller General gave him no alternative but to assess the penalty clause since written application for an extension had not been filed. It took the energetic intervention of a United States senator to get that $6000.

Selection Chart. It is impossible to state flatly which conveyor is or is not best suited to a given material, for conditions vary. Table 2 shows the types usually selected. Of course there are several different conveyors adapted to any one material, and only experience can guide in selecting the best one for a given set of circumstances. The table refers to materials frequently met in material-handling problems. There are many others.

TABLE 1

CHARACTERISTICS OF MATERIALS IN RELATION TO THEIR BEHAVIOR IN CONVEYORS

Note. Power calculations for flight conveyors involve the coefficient of friction of the material in contact with steel plate. In calculating pneumatic conveyors, use the " specific weight," not the weights given in this table.

Material	Weight per Cubic foot	Comment
Acid phosphate	90	Very tricky because sticky. Pneumatic handling impossible.
Alum, lumpy	60–65	Abrasive. Contamination may be objectionable.
Ammonium nitrate	62	Actively hygroscopic. In contact with copper, zinc, or brass it forms a salt which sensitizes it. Not affected by steel or iron. Very sticky when damp. It is an insensitive explosive.
Aluminum oxide	60	One of the most abrasive materials. Rated 9 in Mohs' scale.[1] Fine white powder, 180–300 mesh.

[1]Mohs' scale: 1. Talc. 2. Gypsum. 3. Calc spar. 4. Fluorspar. 5. Apatite. 6. Feldspar. 7. Quartz. 8. Topaz. 9. Sapphire. 10. Diamond.

TABLE 1—*Continued*

Material	Weight per Cubic foot	Comment
Arsenic salt	100–130	Heavy. Poisonous. Compact. Difficult.
Ashes, dry	35–40	Abrasive when dry; corrosive when
wet	45–55	wet.
Bauxite, crushed	75–85	Abrasive.
Bentonite	50	Sluggish.
Bone meal	55–60	Difficult at times.
Borax	50–70	Extremely abrasive.
Bran	16	Easily handled except when oily.
Brewer's grains, wet	55	Sometimes actively corrosive when hot.
Carbon	50	If for certain purposes, as for dry batteries, must be kept from contamination by iron dust, etc.
Carbon black	25	Extremely fragile if pelletized.
Cement	85–100	Entrains air. Compacts in storage. Dusty and abrasive. Must not come in contact with moisture or surfaces which may sweat. En-masse conveyors not advised.
Chalk, pulverized	70–75	Tricky.
Charcoal	12–20	Dusty; dust is explosive.
Clays	35–52	Sluggish. Sometimes abrasive. Adheres to metals.
Coal	40–50	Some are corrosive when wet. Dust is explosive. Degradation may be objectionable. Lignite very fragile.
Cocoa	35–40	Hygienic precautions. Sometimes sticky. Cocoa shells abrasive.
Coffee beans	32–35	Hygienic precautions. Gentle handling.
Copra cake	25–30	Oily. Not difficult.
Copra meal	40	Sticky. Often sandy and abrasive. Oil may exude if handled by pneumatic conveyor.
Cork, ground	5–15	Sluggish. May be discolored by con-
crushed	6–12	tact with iron or steel.
Corn, shelled	45	Easily handled. Abrasive.
Corn meal	40	Hygienic precautions.
Cottonseed	35–40	Will not flow when fuzzy. Hulls and meal easily handled.
Cullet	80–100	(Broken glass, sometimes heated and sprayed with water.) Small particles work into chain joints, causing severe abrasion and sometimes corrosion. Cullet and aluminum oxide are leaders in destructive action on conveyor chains.
Face powder	30–40	Must not be discolored.
Feldspar, crushed	100	Extremely abrasive.

TABLE 1—*Continued*

Material	Weight per Cubic foot	Comment
Flaxseed	45	Abrasive. Free flowing.
Flour	35–40	Hygienic precautions. Explosive dust.
Flue dust (flue dirt)	100	Abrasive.
Fluorspar	100	Abrasive.
Fly ash	30–45	Tricky when aerated. Not abrasive except when mixed with flue dirt.
Foundry sand	90–110	Abrasive. May be sticky.
Fuller's earth, granular	35	Fragile. Slightly abrasive. May be
spent	60	oily and sticky.
Glass batch, soda ash, lime, fine sand		Abrasive.
Glue, granular	45	Must be kept cool and dry.
Graphite, flake	40	Easily handled. Self-lubricating.
flour	25	
Grass seed	11–15	Reasonably gentle handling.
Gypsum, calcined	55–60	Handles readily except when extremely
pulverized	80	fine.
lumpy	90	
Iron ore	150	Abrasive. May be in large fragments.
Lead oxides	60–150	Lead salts are poisonous. Dustless operation essential.
Lead sulphates	100	Abrasive. Dusty.
Lime, pebble	55	Sometimes sluggish.
hydrated	32–40	
pulverized	32–40	
Limestone, dust	85	Abrasive and dusty.
screenings	90–95	Abrasive.
Linseed meal	44	Easily handled.
Malt, dry	20–30	Easily handled.
Mica, pulverized	13–15	Dusty. Somewhat sticky.
Molybdenum concentrates	110	Sticky and difficult. Abrasive.
Oats	25	Shell is fragile. Degradation is objectionable.
Peanuts, shelled	35–40	Hygienic handling. Fragile. Oily.
whole	15–20	Often sandy. Belt elevator best.
Petroleum coke	42	Not particularly abrasive.
Pitch, dry	50–62	Dusty. Lumps fragile.
Pumice	45	Abrasive. Polishes.
Pumicite	40–47	300 mesh, abrasive.
Quartz, ground	110	Extremely abrasive.
Rice	45–50	Hygienic precautions. Fragile.
Rubber scrap, ground	30	Sluggish and difficult.
Salt, coarse	45–50	Hygroscopic. Adheres to iron and
fines	75–80	steel. Difficult when wet and when hot. Corrosive.
Salt cake	75–95	Flows very freely.
Sawdust and pulverized wood	14–20	Floating dust may be explosive.
Shellac	80	Dust dangerously explosive.

TABLE 1—*Continued*

Material	Weight per Cubic foot	Comment
Silica flour	80	Sluggish. Abrasive.
Sludge, sewage	40–50	Sticky, sandy, abrasive. Extremely difficult for any type of conveyor.
Soap flakes (Rinso, etc.)	10–20	Granular. Fragile. Sticky when hot.
Soap powders	30–40	Gentle handling necessary.
Soda ash, heavy	55–65	Caustic. Dustless operation essential.
light	20–35	Sluggish but not difficult. May be as hot as 400° F., which means reduced tension in chains subjected to this heat.
Sodium aluminum sulphate	75	Must not come in contact with oil.
Soybeans, cake	45	Abrasive when sandy. Some of the processed products are sticky, e.g., " expeller foots."
meal	40	
spent	28	
Starch, pulverized	40–45	Dust extremely explosive.
Sugar, raw	55–65	Hygienic requirements. Sometimes sticky. Sheen of refined product must not be impaired.
refined	50–55	
Sulphur, coarse	60–70	Danger from static sparks. Risk of ignition. Corrosive when wet.
pulverized	60	
Talc	50–60	Semi-abrasive. Discoloration objectionable. Adheres to metal.
Tanbark	60	Sluggish. Sometimes corrosive.
Tobacco stems	25	Stringy and sluggish.
Wheat	48	Hygienic conditions as with all food products.
Wood, " hog-fuel "	15–30	Difficult because of slivers, especially in flight and screw conveyors.
Wood chips, dry	18–20	Fluffy when dry and not difficult if screened. Sometimes mildly corrosive.
wet	55	
Zinc and other metallic dusts		Hazardous with en-masse conveyors.
Zinc oxide	20–35	Avoid discoloration. Tends to pack.
Zinc sulphate	60	Not difficult.

TABLE 2

Selection Chart

	Ashes, Wet	Ashes, Dry	Coal, Average	Coal, Corrosive	Coke	Coke Breeze	Flue Dirt	Fly Ash	Grains	Granular and Pulverized Materials	Hog Fuel	Ores, Light	Ores, Heavy	Rock	Sand and Gravel	Sawdust	Sticky Material	Material at High Temperature
Conveyors																		
Apron			*	*	*	*					*	*	*	*				*
Belt			*	*	*	*	*		*	*	*	*	*	*	*	*		Woven wire OK
En masse			*					*	*	*						*		*
Flight			*			*		*		*	*					*		*
Drag chain	*	*		*	*	*		*								*		*
Pneumatic		*	*			*		*		*	*	*				*		*
Vibrating			*	*	*	*				*	*	*	*		*	*		*
Elevators																		
Centrifugal discharge	*	*	*	*			*	*	*	*	*	*			*	*		*
Continuous bucket			*			*		*	*	*	*	*	*	*	*	*	*	*
Gravity discharge			*					*										
Pivoted bucket	*	*	*	*	*	*	*			*								*
Screw conveyors			*			*												
Steel flight			*					*	*	*	*	*				*		*
Cast-iron flight	*	*		*	*		*	*				*						*
Ribbon								*		*	*	*				*	*	*
Steam jet conveyors		*					*	*										
Skip hoists	*	*	*	*	*	*	*	*		*		*		*	*			*
Crushers																		
Roll			*	*	*													
Jaw and gyratory												*	*	*				
Screens																		
Revolving					*	*						*	*	*	*			
Shaker			*	*	*	*					*							
Vibrating			*	*	*	*			*	*	*	*	*	*	*	*		*

CHAPTER 2

SCREW CONVEYORS

The screw conveyor usually consists of a long-pitch plate-steel helix mounted on a shaft or spindle carried in bearings within a U-shaped trough. As the element rotates, the material fed to it is moved forward by the thrust of the lower part of the helix and is discharged through openings in the trough bottom or at the end. This conveyor does not always find favor with engineers, but properly applied it does a good job and often its cost will be about one-half that for any other type of conveyor. It is a simple machine to maintain, replacements are inexpensive, and it is readily made dust tight. For many functions it is the best type of conveyor.

Construction. A well-designed screw conveyor for non-abrasive, non-corrosive materials has a steel helicoid, cold rolled in one continuous strip between bearings, rather than in short sections. By cold rolling the helicoid attains a tapered section with greatest thickness at the shaft and with the thin outer edge taking a hardness about double that of the original strip. If the conveyor is to handle food products, toilet powder, moist cork, or other material which may be discolored through contact with steel plate, the rotating parts or the entire conveyor may be of stainless steel, bronze, aluminum, Monel metal, etc., though at a considerable increase in first cost.

If it is desired to churn or mix materials the helix is made segmental or bladed, like a succession of marine screw propellers, and is then referred to as a paddle mixer. One such application is mixing water with ashes when discharging from a storage bin to reduce the dust nuisance.

For ashes, clinker, ores, etc., the helix may be of cast-iron sections bolted to the shaft and the trough may be made of cast iron or hardened-face concrete. For sticky material, like molasses, hot tar, or asphalt, a ribbon helix is preferable. This construction prevents build-up along the line between helix and shaft where it would cause loss of handling capacity.

Thrust bearings should if possible be located at the discharge end of a conveyor so that the shaft will be in tension. Anti-friction bearings may be provided which take thrust in either direction; they are well

worth the difference in cost over sleeve bearings, as they are in housings with seals to keep out dust and retain the lubricant.

Care should be exercised when ordering a screw to specify whether it is to be right-hand or left-hand. A right-hand screw is one in which the material moves toward you when you stand at the discharge end of the conveyor with the helix rotating clockwise (Fig. 2-1).

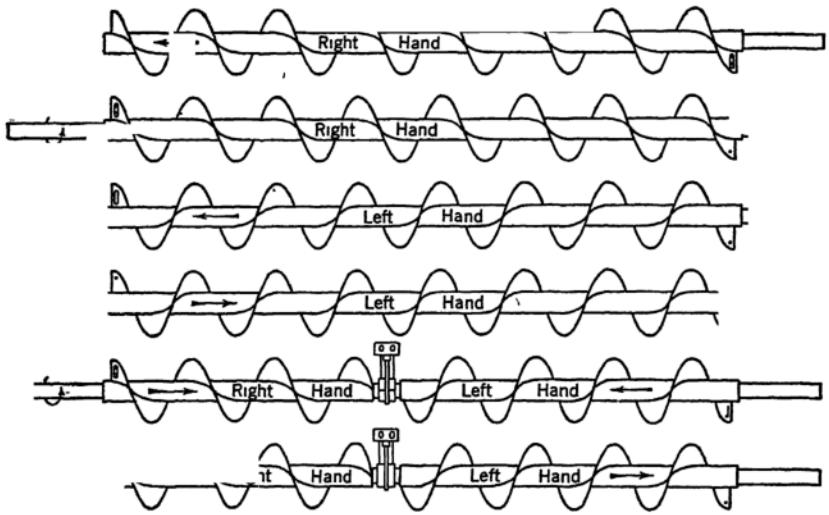


FIG. 2-1. Method of designating screw conveyors as right or left hand.

Inclined Conveyors. Screw conveyors can be operated with the path inclined upward, but the capacity decreases rapidly as the inclination increases (Fig. 2-2). A standard pitch screw inclined at 15° with the horizontal has 70 per cent of its horizontal capacity; if it is inclined 25° the capacity is reduced to 40 per cent; if it is inclined 45° the material will move along the floor of the trough but at a greatly reduced rate. For steep inclines the helix may have a short pitch, and the trough may be made tubular; then the capacity loss is less. Thus with a jam feed and 45° incline such a conveyor has a capacity about 50 per cent of its capacity when horizontal.

The bearing hangers of inclined conveyors should be of T design to reduce the interference with the flow of the material. It may be desirable to use a few turns of double helix each side of the bearing hangers. With suitable forced feed a screw conveyor in a vertical position will lift material if the rotating speed is high and there are no intermediate bearings. Such a conveyor is not self-clearing and should not be used for material which will sour or spoil.

Capacity. Because of the manner in which the material is moved in a screw conveyor the allowable loading and the speed are controlled by the characteristics of the material; thus light free-flowing non-abrasive materials may fill the trough deeply, and the rotating speed may be higher than with heavier and more abrasive materials. The recommendations of the manufacturers should be followed. Those of the Link-Belt Company will be used for illustration.

Figure 2-3*A*. For materials which are light, free flowing, and non-abrasive, such as dry grains, flaxseed, malt, cornmeal, and pulverized coal, the maximum loading is 45 per cent of the vertical cross section

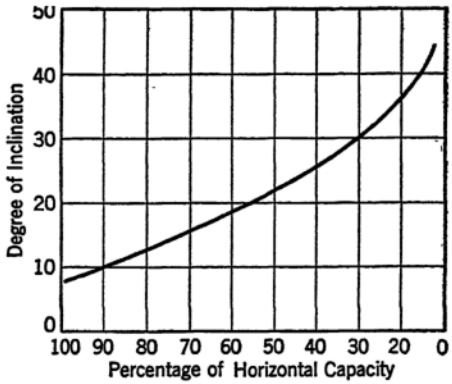


FIG. 2–2. Capacity change as related to inclination with standard pitch conveyors.

of the helix as shown by the diagram on the chart. Maximum speed depends upon the diameter. Thus a 4-in. screw may operate at 180 r.p.m., and 45 per cent loading will give a capacity of 100 cu. ft. per hr. If the diameter is 24 in. (usually the largest standard size), maximum speed is about 95 r.p.m., and 45 per cent loading will give a capacity in excess of 10,000 cu. ft. per hr. Multiplying the number of cubic feet per hour by the weight per cubic foot and dividing by 2000 gives the capacity in tons per hour.

Figure 2-3*B*. For non-abrasive materials of medium weight containing fines and small lumps, such as alum, beans, soybeans, slack coal, hydrated lime, coffee, corn grits, shelled corn, and cottonseed bulbs, the maximum loading is 38 per cent of the vertical cross section of the helix, and the maximum speeds are within the limits of 120 and 70 r.p.m.

Figure 2-3*C*. For semi-abrasive granular materials with small lumps and weighing 40 to 80 lb. per cu. ft., such as lumpy lime, asbestos, borax, carbon black, sized coal, fly ash (not mixed with flue dust), wet

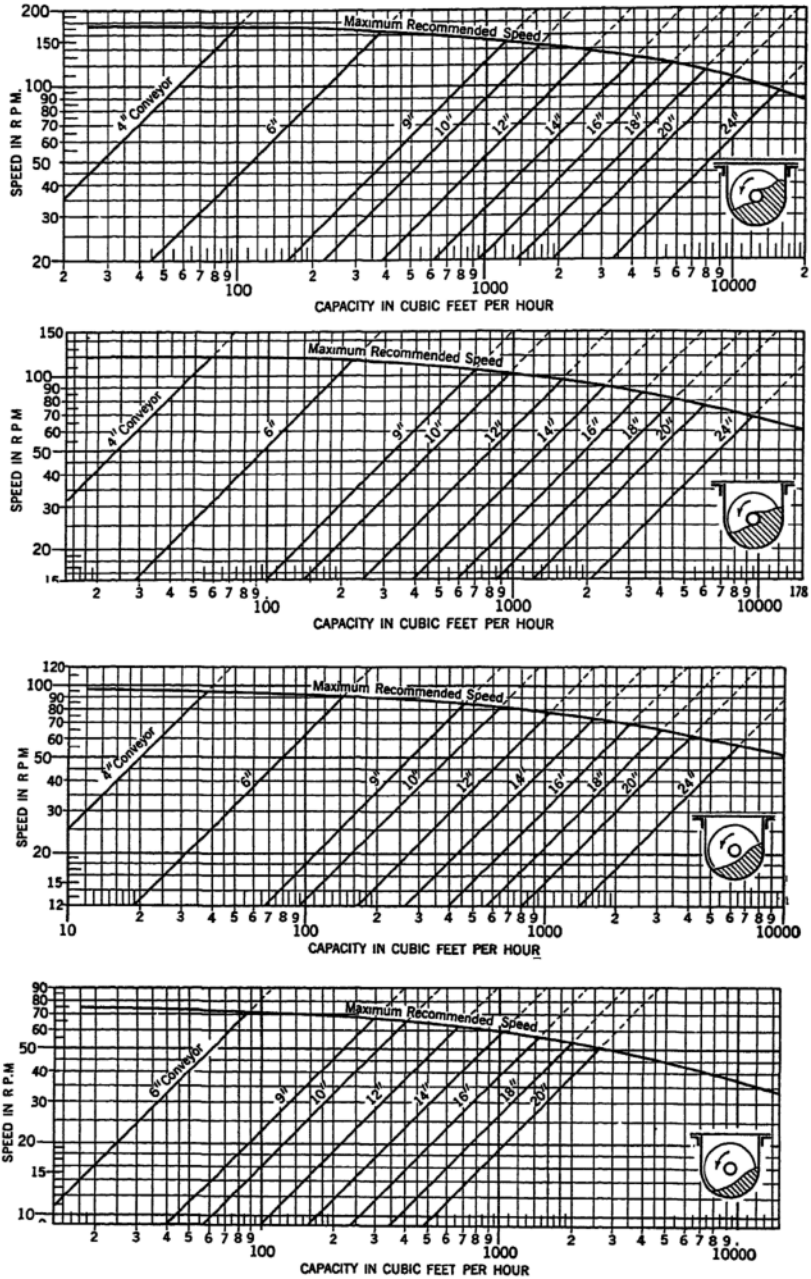


FIG. 2–3. Speed-capacity charts. (Link-Belt.) (*A*) Loading 45 per cent. (*B*) Loading 38 per cent. (*C*) Loading 31 per cent. (*D*) Loading 25 per cent.